



UNITED STATES PATENT AND TRADEMARK OFFICE

20120115891

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/264,586	01/03/2012	Dennis Jones	029369.0000/	9581
4372	7590	10/03/2014		
ARENT FOX LLP 1717 K Street, NW WASHINGTON, DC 20036-5342			EXAMINER NEAGU, IRINA	
			ART UNIT 1627	PAPER NUMBER
			NOTIFICATION DATE 10/03/2014	DELIVERY MODE ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

DCIPDocket@arentfox.com
IPMatters@arentfox.com
Patent-Mail@arentfox.com

Office Action Summary

Application No.
13/264,586

Applicant(s)
JONES, DENNIS

Examiner
IRINA NEAGU

Art Unit
1627

AIA (First Inventor to File)
Status
No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.

If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.

Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133).

Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

- 1) ☒ Responsive to communication(s) filed on 25 August 2014.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.
- 2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.
- 3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.
- 4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

- 5) ☒ Claim(s) 1, 11-14 and 16-20 is/are pending in the application.
5a) Of the above claim(s) 14 and 16-20 is/are withdrawn from consideration.
- 6) ☐ Claim(s) ____ is/are allowed.
- 7) ☒ Claim(s) 1, 11-13 is/are rejected.
- 8) ☐ Claim(s) ____ is/are objected to.
- 9) ☐ Claim(s) ____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

- 10) ☐ The specification is objected to by the Examiner.
- 11) ☐ The drawing(s) filed on ____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

- 12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some** c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. ____.
3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

- 1) ☒ Notice of References Cited (PTO-892)
- 2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date: ____
- 3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date: ____
- 4) ☐ Other: ____

Art Unit: 1627

The present application is being examined under the pre-AIA first to invent provisions.

DETAILED ACTION

Claims 1, 11-14 and 16-20 are pending in the instant application.

Claims 14 and 16-20 are withdrawn, as being drawn to a non-elected invention.

Claims 1 and 11-13 are being examined herein.

Response to Arguments of 25 August 2014

The rejection of claims 1, 1 and 11-13 under 35 U.S.C. 101 has been reconsidered and is herein withdrawn.

Applicant's Declaration under 37 C.F.R. 1.131, submitted on 25 August 2014, in which inventor Dennis Jones stated on record that he had reduced to practice "at least as much what Mukherjee is alleged by the Examiner to disclose" prior to Mukherjee's publication date, is acknowledged. In view of Applicant's Declaration of 25 August 2014, the reference by Mukherjee is disqualified, and therefore the rejection of claims 1 and 11-13 under 35 U.S.C. 103(a) over Uccatadyamodakah, Van Der Glessen, Nilakamala Kanda Gunah, and Mukherjee, in further view of Jones, Maletto and Chinery; and the rejection of claims 1 and 11-13 under 35 U.S.C. 103(a) over Ram Hart and Mukherjee, in view of Jones, Maletto and Chinery are herein withdrawn.

The examiner wishes to clarify the following: Mukherjee was used in the rejections of 25 October 2013 and again on 23 April 2014 for the teachings that:

1. *Nelumbo nucifera* extract has anti-obesity activity and
2. *Nelumbo nucifera* inherently contains alkaloids such as nuciferine.

None of these teachings by Mukherjee were new at the time of the instant invention, rather the antiobesity activity of *Nelumbo nucifera* has been known since 2000 (extracts of *Nelumbo* had been used to treat obesity in China), and the inherent content of nuciferine in *Nelumbo nucifera* has been known since 1959. Furthermore, the examiner notes that a method of inducing a feeling of having already eaten, or of stimulating in a subject a feeling of satiation with a composition comprising *Mucuna pruriens* has been known for more than 2000 years, and a method of inducing a feeling of having already eaten, or of stimulating in a subject a feeling of satiation with *Nelumbo nucifera* has been known since 500 B.C.; these methods render obvious the instantly claimed method of inducing a feeling of having already eaten, or of stimulating in a subject a feeling of satiation with a combination of *Mucuna pruriens* and *Nelumbo nucifera*. Thus, while the reference by Mukherjee is herein replaced with a different reference, and modified rejections under 103(a) are made below, the grounds for the rejection made herein are effectively the same as in the non-final rejection of 23 April 2014. The examiner considers Applicant's arguments of 25 August 2014 to be relevant to the modified rejections under 35 U.S.C. 103(a) below.

Applicant's arguments (Remarks of 25 August 2014, pages 6-15) regarding the rejection of claims 1 and 11 under 35 U.S.C. 103(a) over Uccatadyamodakah and Nilakamala Kanda Gunah, in view of Van Der Glessen, and the rejection of claim 12 and 13 in further view of Jones, Maletto and Chinery, have been carefully considered, but are not found persuasive at least for the reasons below:

Applicant argues (page 8) that the examiner has misinterpreted the cited art.

Art Unit: 1627

Applicant argues (page 8, first paragraph) that Uccatadyamodakah teaches a combination product containing 11 different ingredients, one of which is *Mucuna pruriens*, in a “bolus like sweet preparation”. The examiner points out that the language of the instant claims “a composition comprising” is inclusive of additional ingredients, in a composition of *Mucuna pruriens*.

In response to Applicant’s statement (page 8, first paragraph): “Applicants do not understand how 48 grams of the above combination [of 11 ingredients] can equate with 50 mg to 500 mg of L-DOPA,” and the argument (page 8, second paragraph) that “the examiner fails to explain how one arrives from the information of Van Der Giessen to the amounts of L-DOPA present in what Uccatadyamodakah teaches”, the examiner maintains that Uccatadyamodakah is only used for teaching a method of stimulating in a subject a feeling of satiation with a therapeutic formulation comprising *Mucuna pruriens*, while Van Der Giessen (US 2006/0165822), is used in the rejection for the teaching that the L-Dopa content is an inherent property of the seeds of *Mucuna pruriens*. Specifically, Van Der Giessen teaches an L-Dopa content of 4.86% or 250 mg per sachet ([0107]) of *Mucuna* seed powder (each sachet containing 7.5 grams), which is within the range of instant claim 1. Applicant’s argument (page 8, second paragraph) that “the sachet taught by Van Der Giessen is not necessarily a dosage form but a unit of the plant material provided” is irrelevant. Thus, Van Der Giessen teaches not only the use of *Mucuna pruriens*, but also the use of the very same dose/amount of *Mucuna pruriens* as in instant claim 1.

In response to Applicant’s argument (page 9) that Uccatadyamodakah teaches a number of conditions treated with *Mucuna pruriens*, besides satiating, the examiner emphasizes that the

Art Unit: 1627

fact that *Mucuna* can be used to treat other conditions does not diminish in any way the teaching by Uccatadyamodakah that *Mucuna* or an extract thereof is useful for appetite suppression or satiating. Further, the examiner notes that administering the very same composition, in this case a composition comprising *Mucuna pruriens*, to the same population will inherently result in the same effect.

In response to Applicant's argument (page 9, last paragraph-page 10, first paragraph) that the examiner's reliance on Van Der Giessen is flawed, the examiner maintains that Van Der Giessen teaches administration of a composition comprising *Mucuna pruriens*, wherein the amount of L-DOPA is 250 mg, as in instant claim 1. Administration of the same product to the same population will inherently produce the same result. "Products of identical chemical composition can not have mutually exclusive properties." A chemical composition and its properties are inseparable. Therefore, if the prior art teaches the identical chemical structure, the properties applicant discloses and/or claims are necessarily present. *In re Spada*, 911 F.2d 705, 709, 15 USPQ2d 1655, 1658 (Fed. Cir. 1990) (Applicant argued that the claimed composition was a pressure sensitive adhesive containing a tacky polymer while the product of the reference was hard and abrasion resistant. "The Board correctly found that the virtual identity of monomers and procedures sufficed to support a *prima facie* case of unpatentability of Spada's polymer latexes for lack of novelty."). See 2112.01.

Applicant argues (page 10 last paragraph, page 11 first paragraph): "Although Uccatadyamodakah teaches taking this sugar bolus [*Mucuna* preparation with honey and sugar] on the empty stomach in the evening, Uccatadyamodakah does not suggest taking this before any meal". The examiner does not see the difference between administration "on the empty stomach"

Art Unit: 1627

and "before a meal". Applicant argues for "a more logical conclusion" that the sugar bolus is a meal substitute. The examiner does not agree with Applicant's proposed "more logical conclusion" and finds the argument speculative and irrelevant. If Applicant believes that the novelty or non-obviousness of the invention lies in the fact that the instant composition is administered exactly 45 minutes before a meal, Applicant should provide support in the form of unexpected results for such a claim. In the absence of such showing of unexpected results, the examiner maintains that deciding when a composition is administered, before a meal, on an empty stomach, etc., is well within the skill of any artisan, and therefore prima facie obvious.

Based on all the arguments above, the examiner disagrees with Applicant's assertion that the examiner has misinterpreted the prior art and maintains all the reasons for the obviousness rejection under 35 U.S.C. 103(a).

Applicant's argument (page 11) that

2. No Motivation to Combine

In view of the deficiencies in the Examiner's interpretation of Uccatadyamodakah and Van Der Giessen, applicants assert that the Examiner's rationale for combining these references with the teachings of Nilakamala Kanda Gunah and Mukherjee fails. That is, the Examiner has not established that Uccatadyamodakah and Van Der Giessen, either alone or in combination, suggests a method of inducing in a subject a feeling of having already eaten using *Mucuna pruriens*. Thus, the Examiner cannot say that motivation to combine stems from the knowledge that both *Mucuna pruriens* and *Nelumbo nucifera* were known to induce satiation.

is not

found persuasive. The examiner has clearly established that Uccatadyamodakah teaches a method to induce satiation with *Mucuna pruriens*, and Van Der Giessen teaches administration of *Mucuna pruriens* to a subject in an amount corresponding to 250 mg of L-DOPA, as in instant

Art Unit: 1627

claim 1. Administering the very same composition, in this case a composition comprising *Mucuna pruriens*, to the same population will inherently result in the same effect.

In response to Applicant's argument regarding the Rule 131 Declaration, the examiner indicates that the reference by Mukherjee has been disqualified. The reference by Mukherjee has been replaced with another reference, while maintaining the very same grounds of rejection, and the modified rejection is shown below.

Applicants have set forth no new arguments (page 12) against the rejection of claims 12 and 13 over Uccatadyamodakah, Van Der Glessen, Nilakamala Kanda Gunah, and Mukherjee, in further view of Jones, Maletto and Chinery.

Applicant's arguments against the rejection of claims 1 and 11-13 under 35 U.S.C. 103(a) over Ram Hart and Mukherjee, in view of Jones, Maletto and Chinery have been considered, but are not found persuasive.

Applicant argues (page 14) that Hart teaches "a specific formulation that removes time constraints around meals" and thus "teaches against administration around food intake". While the examiner acknowledges that Hart teaches buccal/ sublingual delivery, the examiner is not persuaded by Applicant's argument that Hart teaches against administration around food intake. There is nothing in the reference by Hart that teaches that food cannot be eaten after consuming the dietary supplement. On the contrary, if the lozenge is administered before subjects consume any food, the fact that food will be consumed after taking the lozenge is implicit. Further, Hart specifically teaches that "lozenges are dissolved in the mouth every three to four hours" ([0080],

Art Unit: 1627

lines 2-3). Thus, Hurt implicitly teaches that the dietary supplement is administered at some time before food intake.

Applicant argues (page 13, last paragraph) that Hurt is not teaching or suggesting a method of stimulating a feeling of having already eaten. The examiner does not find such an argument persuasive for the following reasons: 1. as previously indicated, Hurt specifically teaches administration of the composition before eating; no food was permitted until completion of the study [0084]; and 2. as previously indicated, Hurt teaches a method of regulating hunger [0171] with *Mucuna pruriens*. Further, Hurt specifically teaches that the dietary supplement comprising *Mucuna pruriens* improves eating satisfaction ([0099], line 4), and has improved more than ten times eating satiation ([0097], last line).

Thus, for all the reasons presented above, the rejection of claims 1 and 11-13 is maintained and modified rejections under 35 U.S.C. 103(a) are made below.

Upon further search and consideration, as well as careful review of the references submitted in the IDS, it has been determined that the presently submitted claims are subject to the following grounds of rejections under 35 U.S.C. §103. The rejections are detailed below. Accordingly, this action is non-final.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103(a) which forms the basis for all obviousness rejections set forth in this Office action:

(a) A patent may not be obtained though the invention is not identically disclosed or described as set forth in section 102 of this title, if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject

Art Unit: 1627

matter pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103(a) are summarized as follows:

1. Determining the scope and contents of the prior art.
2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

Claims 1 and 11 are rejected under 35 U.S.C. 103(a) as being unpatentable over Uccatadyamodakah (In *Vangasena*; Shankar et al., Eds.; Bombay, India, 1996; p. 236; cited as Exhibit 1 in the IDS of 29 March 2013), Van Der Glessen et al. (US 2006/0165822, published 27 July 2006, cited in the IDS of 14 October 2011), Nilakamala Kanda Gunah (Kaiyadeva, Nilakamala Kanda Gunah. In *Kaiyadevanighantau*; Sharma et al., Eds.; Varanasi, India, 1979; p. 268-269; cited as Exhibit 3 in the IDS of 29 March 2013), Ono et al. (J. Ethnopharmacology 2006, 106, 238-244, cited in PTO-892) and Shamma et al. (Chem. Rev. 1964, 64 (1), 59-79, cited in PTO-892).

Uccatadyamodakah teaches a method of stimulating in a subject a feeling of satiation with a therapeutic formulation comprising *Mucuna pruriens*. Uccatadyamodakah teaches the administration is oral, in the evening, on empty stomach (before the subject eats a meal, as in claims 1, 11).

Uccatadyamodakah does not teach a method of stimulating a feeling of having already eaten with a combination of an extract from *Mucuna pruriens* and *Nelumbo nucifera*.

Art Unit: 1627

Uccatadyamodakah does not specifically teach administration of the composition 45 minutes before the subject eats a meal, as in instant claim 11.

Even though Uccatadyamodakah does not specifically teach that an extract of *Mucuna pruriens* contains L-Dopa, the L-Dopa content is an inherent property of the seeds of *Mucuna pruriens*, known at the time the invention was made, as evidenced by Van Der Glessen et al. (US 2006/0165822, published 27 July 2006, cited in the IDS of 14 October 2011, see Abstract, [0016], *Mucuna pruriens* used either as seed powder or as extract, also [0022]). Specifically, Van Der Glessen teaches an L-Dopa content of 4.86% or 250 mg per sachet ([0107]) of *Mucuna* seed powder (each sachet containing 7.5 grams), which is within the range of instant claim 1.

Moreover, Van Der Glessen teaches that *Mucuna pruriens*, or an extract thereof, is used to treat metabolic and nutritional disorders ([0058]).

Nilakamala Kanda Gunah teaches a method of stimulating in a subject a feeling of satiation with a therapeutic formulation comprising *Nelumbo Nucifera*.

Ono et al. (J. Ethnopharmacology 2006, 106, 238-244, cited in PTO-892) teaches that an extract from *Nelumbo nucifera*, which contains several alkaloids (page 238, left column, last line), is used to treat obesity (Abstract, also page 238, right column, lines 2-4).

Even though Nilakamala Kanda Gunah and Ono do not specifically teach that an extract of *Nelumbo nucifera* contains an aporphine alkaloid such as nuciferine, the nuciferine content is an inherent property of the *Nelumbo nucifera* seed extract, known at the time the invention was made, as evidenced by Shamma et al. (Chem. Rev. 1964, 64 (1), 59-79, cited in PTO-892).

Art Unit: 1627

Shamma teaches that *Nelumbo nucifera* contains nuciferine, an aporphine alkaloid (page 67, left column, lines 12-13).

It would have been obvious to a person of ordinary skill in the art at the time the invention was made to combine an extract of *Mucuna pruriens* and an extract of *Nelumbo nucifera*, in a therapeutic formulation, wherein the amount of L-Dopa is 50 mg to 500 mg and the amount of nuciferine alkaloid is 2 mg to 50 mg, and use the formulation to stimulate a feeling of having already eaten.

The person of ordinary skill in the art at the time the invention was made would have been motivated to combine an extract of *Mucuna pruriens* and an extract of *Nelumbo nucifera*, because *Mucuna pruriens* (seeds or extract) was known to induce satiation, as taught by Uccatadyamodakah, and was known to be used (in doses/ sachets of seeds containing 250 mg L-Dopa) to treat metabolic and nutritional disorders, as taught by Van Der Glessen; and

Nelumbo nucifera was known to induce a feeling of satiation in a subject and was known to have anti-obesity activity, as taught by Nilakamala Kanda Gunah and Ono, and was known to contain nuciferine alkaloid, as taught by Shamma. Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Mucuna pruriens* and *Nelumbo nucifera*, known to be useful for the same purpose, and optimizing the ratio between the active ingredients, will result in an appetite-suppressing composition. Such an optimization of the ratio between the content of *Mucuna pruriens* and *Nelumbo nucifera* in the formulation (which corresponds to the claimed content of L-Dopa versus nuciferine alkaloid), with the aim of maximizing the therapeutic effect, is well within the competency of a person of ordinary skill in

Art Unit: 1627

the art. Since all composition components herein are known to be useful as inducing satiation, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

Furthermore, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to administer the combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* 45 minutes before the subject eats a meal. The person of ordinary skill in the art would have been motivated to administer the composition before the subject eats a meal, because compositions comprising *Mucuna pruriens* were known to be administered orally on an empty stomach, as taught by Uccatadyamodakah. Varying the time before the composition is administered to the subject with the aim of improving the therapeutic effect of the composition is considered routine for the skilled artisan.

As such, claims 1 and 11 are rejected as *prima facie* obvious.

Claims 12 and 13 are rejected under 35 U.S.C. 103(a) as being unpatentable over Uccatadyamodakah, Van Der Glessen, Nilakamala Kanda Gunah, Ono and Shamma, as applied to claims 1 and 11 above, in further view of Jones (US 6,224,873 of 1 May 2001, cited in PTO-892 of 25 October 2013), Maletto et al. (US 2006/0078627, published 13 April 2006, cited in PTO-892 of 25 October 2013) and Chinery (US 2004/0077556, published 22 April 2004, cited in the IDS of 14 October 2011).

Uccatadyamodakah, Van Der Glessen, Nilakamala Kanda Gunah, Ono and Shamma are as above.

Jones (US 6,224,873) teaches a method of weight control/loss by regulation of appetite using compositions comprising *Citrus aurantium* (Examples 2, 3, 5, columns 17-18, subjects completed a daily appetite and satiety rating questionnaire). Jones teaches (column 16, last paragraph) that the Citrus material, either in the form of an extract or as the natural material, may be given in combination with other herbs that possess beneficial effects for humans, particularly in respect to weight loss.

Maletto (US 2006/0078627) teaches (page 2, [0025], [0027]) a method of reducing hunger with a composition comprising *Griffonia simplicifolia*. Maletto teaches that *Griffonia simplicifolia* is a natural source of 5-hydroxytryptophan, which is a precursor to serotonin ([0027]). Maletto teaches that serotonin is one of the most important brain neurotransmitters involved in appetite suppression; it has been established that increased serotonin levels decrease appetite.

Chinery (US 2004/0077556) teaches a method for promoting weight loss in mammals with a nutritional supplement composition comprising *Camellia sinensis* ([0006]). Chinery teaches that green tea extract, *Camellia sinensis*, contains epigallocatechin gallate (EGCG), which suppresses appetite ([0006]).

It would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Citrus aurantium* and an extract from *Griffonia simplicifolia* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera*, in a therapeutic formulation used to stimulate a feeling of having already eaten.

The person of ordinary skill in the art at the time the invention was made would have been motivated to to add an extract from *Citrus aurantium* and an extract from *Griffonia*

Art Unit: 1627

simplicifolia to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* and use the combination in a method of stimulating in a subject a feeling of satiation, because extracts of *Citrus aurantium* were known to suppress appetite and induce weight loss, as taught by Jones, and extracts of *Griffonia simplicifolia* were known to reduce hunger, as taught by Maletto. The person of ordinary skill in the art would have been further motivated to add such extracts together because Jones teaches that *Citrus aurantium*, either in the form of an extract or as the natural material, may be given in combination with other herbs that possess beneficial effects for humans, particularly in respect to weight loss.

Similarly, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Camellia sinensis* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera*, in a therapeutic formulation used to stimulate a feeling of having already eaten. The person of ordinary skill in the art at the time the invention was made would have been motivated to add an extract from *Camellia sinensis* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* and use the combination in a method of stimulating in a subject a feeling of satiation, because extracts of *Camellia sinensis* were known to suppress appetite and promote weight loss, as taught by Chinery.

Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Citrus aurantium*, *Griffonia simplicifolia*, *Mucuna pruriens* and *Nelumbo nucifera*, or combining extracts from *Camellia sinensis*, *Mucuna pruriens* and *Nelumbo nucifera*, known to be useful for the same purpose, will result in an appetite-suppressing composition. Since all composition components herein are known to be useful as inducing satiation, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At

Art Unit: 1627

least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

As such, claims 12 and 13 are rejected as *prima facie* obvious.

Claims 1 and 11-13 are rejected under 35 U.S.C. 103(a) as being unpatentable over Ram Hart et al. (US 2006/0030625, published 9 February 2006, cited in PTO-892 of 23 April 2014), Ono et al. (J. Ethnopharmacology 2006, 106, 238-244, cited in PTO-892) and Shamma et al. (Chem. Rev. 1964, 64 (1), 59-79, cited in PTO-892), in view of Jones (US 6,224,873 of 1 May 2001, cited in PTO-892 of 25 October 2013), Maletto et al. (US 2006/0078627, published 13 April 2006, cited in PTO-892 of 25 October 2013) and Chinery (US 2004/0077556, published 22 April 2004, cited in the IDS of 14 October 2011).

Ram Hart et al. (US 2006/0030625) teaches a method of elevating serotonin and dopamine levels in a subject by administering to a subject a composition comprising 390 mg of L-DOPA as velvet bean, *Mucuna pruriens* seed extract, and 5-hydroxytryptophan (5-HTP) as *Griffonia simplicifolia* seed extract ([0081], Example 1, pages 6-7). Further, Hart specifically teaches that said dietary supplement improves eating satisfaction ([0099], line 4), and has improved more than ten times eating satiation ([0097], last line).

Ram Hart also teaches a method of elevating dopamine levels by administering to a subject a composition comprising 175 mg L-DOPA as velvet bean, *Mucuna pruriens* seed extract ([0082], Example 2, page 7). Ram Hart teaches that dopamine is necessary, for example, for hunger regulation ([0017]). Ram Hart teaches administration of said composition before eating (no food was permitted until completion of the study [0084]).

Art Unit: 1627

Ram Hart does not teach a method of stimulating a feeling of having already eaten with a combination of an extract from *Mucuna pruriens* and *Nelumbo nucifera*.

Ono teaches that *Nelumbo nucifera* extract has anti-obesity activity (Abstract, also page 238, right column, lines 2-4).

Shamma teaches that *Nelumbo nucifera* contains the aporphine alkaloid nuciferine (page 67, left column, lines 12-13). Thus, Shamma teaches that the nuciferine content is an inherent property of the *Nelumbo nucifera*.

Jones (US 6,224,873) teaches a method of weight control/loss by regulation of appetite using compositions comprising *Citrus aurantium* (Examples 2, 3, 5, columns 17-18, subjects completed a daily appetite and satiety rating questionnaire). Jones teaches (column 16, last paragraph) that the Citrus material, either in the form of an extract or as the natural material, may be given in combination with other herbs that possess beneficial effects for humans, particularly in respect to weight loss.

Maletto (US 2006/0078627) teaches (page 2, [0025], [0027]) a method of reducing hunger with a composition comprising *Griffonia simplicifolia*. Maletto teaches that *Griffonia simplicifolia* is a natural source of 5-hydroxytryptophan, which is a precursor to serotonin ([0027]). Maletto teaches that serotonin is one of the most important brain neurotransmitters involved in appetite suppression; it has been established that increased serotonin levels decrease appetite.

Chinery (US 2004/0077556) teaches a method for promoting weight loss in mammals with a nutritional supplement composition comprising *Camellia sinensis* ([0006]). Chinery

Art Unit: 1627

teaches that green tea extract, *Camellia sinensis*, contains epigallocatechin gallate (EGCG), which suppresses appetite ([0006]).

It would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract of *Nelumbo nucifera* to a composition taught by Ram Hart, and use the formulation to stimulate a feeling of having already eaten.

The person of ordinary skill in the art at the time the invention was made would have been motivated to add an extract of *Nelumbo nucifera* to a composition taught by Ram Hart, because *Nelumbo nucifera* was known to have anti-obesity activity, as taught by Ono, and a composition comprising 390 mg of L-DOPA as *Mucuna pruriens* seed extract, and 5-hydroxytryptophan (5-HTP) as *Griffonia simplicifolia* seed extract was taught by Ram Hart to improve eating satisfaction and to improve more than ten times eating satiation by elevating serotonin and dopamine levels in a subject. Furthermore, since dopamine levels were known to be necessary in hunger regulation, as taught by Ram Hart, and increased serotonin levels were known to be correlated with decreased appetite, as taught by Maletto, a composition of *Mucuna pruriens* and *Griffonia simplicifolia* taught by Ram Hart was expected to suppress appetite and thus have anti-obesity activity.

Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Mucuna pruriens*, *Griffonia simplicifolia*, and *Nelumbo nucifera*, known to be useful for the same purpose, and optimizing the ratio between the active ingredients, will result in an appetite-suppressing composition. Such an optimization of the ratio between the content of *Mucuna pruriens* and *Nelumbo nucifera* in the formulation (which corresponds to the claimed content of L-Dopa versus nuciferine alkaloid), with the aim of maximizing the therapeutic effect,

Art Unit: 1627

is well within the competency of a person of ordinary skill in the art. Since all composition components herein are known to be useful as inducing satiation, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

Furthermore, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to administer the combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera* 45 minutes before the subject eats a meal. The person of ordinary skill in the art would have been motivated to administer the composition before the subject eats a meal, because compositions comprising *Mucuna pruriens* were known to be administered orally on an empty stomach, as taught by Ram Hart. Varying the time before the composition is administered to the subject with the aim of improving the therapeutic effect of the composition is considered routine for the skilled artisan.

It would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Citrus aurantium* to a combination of extracts of *Mucuna pruriens*, *Nelumbo nucifera* and *Griffonia simplicifolia*, in a therapeutic formulation used to stimulate a feeling of having already eaten. The person of ordinary skill in the art at the time the invention was made would have been motivated to to add an extract from *Citrus aurantium* to the combination and use the combination in a method of stimulating in a subject a feeling of satiation, because extracts of *Citrus aurantium* were known to suppress appetite and induce weight loss, either alone or in combination with other herbs that promote weight loss, as taught by Jones.

Art Unit: 1627

Similarly, it would have been obvious to a person of ordinary skill in the art at the time the invention was made to add an extract from *Camellia sinensis* to a combination of extracts of *Mucuna pruriens* and *Nelumbo nucifera*, in a therapeutic formulation used to stimulate a feeling of having already eaten. The person of ordinary skill in the art at the time the invention was made would have been motivated to add an extract from *Camellia sinensis* to the combination, because extracts of *Camellia sinensis* were known to suppress appetite and promote weight loss, as taught by Chinery.

Therefore, one of ordinary skill in the art would have reasonably expected that combining extracts from *Citrus aurantium*, *Griffonia simplicifolia*, *Mucuna pruriens* and *Nelumbo nucifera*, or combining extracts from *Camellia sinensis*, *Mucuna pruriens* and *Nelumbo nucifera*, known to be useful for the same purpose, will result in an appetite-suppressing composition. Since all composition components herein are known to be useful in suppressing appetite, it is considered *prima facie* obvious to combine them into a single composition useful for the same purpose. At least additive effects would have been reasonably expected. See *In re Kerkhoven*, 205 USPQ 1069 (CCPA 1980).

As such, claims 1 and 11-13 are rejected as *prima facie* obvious.

Conclusion

Claims 1 and 11-13 are rejected.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to IRINA NEAGU whose telephone number is (571)270-5908. The examiner can normally be reached on Monday-Thursday 7:30-5:00.

Art Unit: 1627

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, SREENIVASAN PADMANABHAN can be reached on (571)272-0629. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/IRINA NEAGU/
Examiner, Art Unit 1627

/SREENI PADMANABHAN/
Supervisory Patent Examiner, Art Unit 1627



Search Notes 	Application/Control No. 13264586	Applicant(s)/Patent Under Reexamination JONES, DENNIS
	Examiner IRINA NEAGU	Art Unit 1627

CPC- SEARCHED						
Symbol					Date	Examiner
A61K2300/00	A61K36/752	A61K36/48	A61K36/185	A61K36/82	8 April 2014	IN
A23L1/3002	A23V2250/21	A23V2200/30	Y10S514/909		8 April 2014	IN

CPC COMBINATION SETS - SEARCHED		
Symbol	Date	Examiner

US CLASSIFICATION SEARCHED			
Class	Subclass	Date	Examiner

SEARCH NOTES		
Search Notes	Date	Examiner
EAST inventor search, assignee search, word/term search	10 October 2013	IN
Traditional Knowledge Digital Library (TKDL) search	10 October 2013	IN
EAST inventor search, assignee search, CPC search, word/term search	8 April 2014	IN
EAST inventor search, assignee search, word/term search	29 September 2014	IN

INTERFERENCE SEARCH			
US Class/ CPC Symbol	US Subclass / CPC Group	Date	Examiner

--	--